When Confidence Survives Missing Evidence: A Locked Study of Harmful Audio–Visual Switching

Anonymous research manuscript

Abstract

Post-fusion selectors can remain confident when visual evidence becomes unusable. We investigate this failure through an inspected CREMA-D study and a separately locked external RAVDESS evaluation. On CREMA-D, an earlier clean-tuned threshold changes 53.4% of predictions under a central visual occluder and loses 22.93 accuracy points relative to audio. Requiring estimated positive gain does not repair this operating point: its harmful switches already receive positive scores. A calibration audit traces twelve near-zero rescue estimates to an isotonic zero plateau supported by six non-neutral calibration cases. We implement temporal and crop-consistency probes, equal-information gain and outcome models, less flexible calibration, and support-aware abstention. The new protocol freezes these choices before inference on 576 clips from 24 RAVDESS actors and repeats the evaluation with two expert checkpoints. At the primary fixed-threshold operating point, support-aware routing gains 0.00 points over audio; its paired difference from the margin gate is -4.17 points, with a 95% actor interval [-6.42, -1.91]. The support gate rejects every external clip, including clean video; its zero harmful-switch rate is therefore complete abstention, not selective robustness. The locked primary superiority requirement is not met. We release measured decisions and source-level intervals with batch re-ranking. An exploratory extension evaluates all 169 actor subsets per checkpoint and repairs a sample-dependent certificate region. The evidence supports an operational reliability audit, not a new general oracle-headroom theorem.

1. Introduction

A fusion model is useful only when its additional information improves the decision. A selector that overrides audio after seeing audio and fusion outputs faces a second problem: it must distinguish beneficial corrections from confidently harmful changes. A large outcome-oracle gap does not establish that observable scores can make this distinction. That point is already explicit in audio–visual complementarity and routing diagnostics [2, 3]. Here we ask a narrower deployment question: what happens to a clean-trained selector when the visual stream becomes unreliable?

A batch comparison can understate the practical failure. In the inspected CREMA-D experiment, a 20% batch policy loses 7.32 points under occlusion, but a clean-tuned independent threshold loses 22.93 points. The latter changes more than half the predictions. Both numbers describe the same frozen models and corrupted clips, under different decision rules. Reporting only the bounded batch result hides the severity of independent deployment.

We test three interventions separately: permit fewer changes than the budget, replace flexible calibration, and measure visual evidence beyond final class scores. The first is an operational correction. The second probes a small-support calibration failure. The third tests whether reliability information helps a direct learner as well as structured heads. None is credited as successful merely because it improves on audio in a reused sample.

The study contributes an executable diagnosis, a controlled comparison of remedies, and a newly reserved external test. The central claim is tied to the measured operational behavior and source-level uncertainty. The original inspected test remains development evidence; it is never relabeled as fresh confirmation.

2. Positioning and decision problem

2.1. Relevant prior work

Hurt examines oracle and realized complementarity for frozen audio–visual emotion models [2]. Shihab et al. separate outcome-oracle opportunity, signal-restricted value, and fitted-policy performance [3]. Weighted Bayes overlap [4] supplies the underlying decision-theoretic interpretation. Our study does not introduce this distinction, nor claim novelty for decomposing a signed correction into rescue and harm.

Visual reliability under corruption is also established research. Watch or Listen models visual corruption and learns reliability scores for audio–visual speech recognition [28]. A broader taxonomy separates noisy, incomplete, imbalanced, and varying-quality modalities [29]. Our frozen-head experiment differs in task and decision timing: both experts have already executed, and the question is whether to replace the audio answer. It does not reproduce or outperform an end-to-end speech-recognition system.

Calibration improves the interpretation of probabilities on a reference distribution [9, 23, 33]; it does not automatically establish reliability under dataset shift [30]. Maximum-softmax and perturbation-based detectors [10, 34], Bayesian approximations and ensembles [12–14], and post-hoc confidence repair [32] motivate strong score-based alternatives. We compare decision outcomes at the selected threshold, because global calibration summaries need not describe the region where the policy acts.

Selective classification [11, 31, 35] provides the appropriate option to abstain. Conformal prediction and risk control state their validity assumptions explicitly [20–22]; weighted conformal methods address particular covariate shifts [36]. Our support detector is not such a certificate. We retain a separate finite-region bound precisely to show where a formal clean-population statement becomes too conservative to support useful actions.

Audio–visual representations range from Siamese and equivariant objectives [5, 6] to masked multimodal learning [7] and attention bottlenecks [17]. VGGSound and VGGSounder provide event-level data and modality-aware labels [1, 8]. Earlier work documents failures of multimodal optimization and balance [15, 16]; these are distinct from post-inference routing. SoundNet [24], WavLM [25], DINOv2 [26], and the transformer/ViT architecture [18, 19] contextualize the frozen feature pipeline. CREMA-D [27] and RAVDESS [38] define the two acted-emotion domains here. WILDS [37] motivates treating source shift as an evaluation design problem rather than equating synthetic corruption with natural variation.

2.2. Estimand and accounting

Let $C_A, C_F \in \{0, 1\}$ be correctness of audio and fusion, $\Delta = C_F - C_A$, and D their prediction disagreement. A policy a switches to fusion when $a = 1$ and otherwise retains audio. Its gain is $G(a) = \mathbb{E}[a\Delta]$. The same two-route oracle denominator is $\Pr(\Delta = 1)$ throughout. Video-only is an endpoint, not an additional oracle route.

If $v = \Pr(a = 1)$ and $u = \mathbb{E}[\Delta \mid a = 1]$, then $G = vu$. Rescue, harm, and neutral switches are reported separately; neutral changes consume budget without changing correctness. Comparing conditions 0 and 1 gives the symmetric identity

$$G_1 - G_0 = \tfrac{u_1+u_0}{2}(v_1 - v_0) + \tfrac{v_1+v_0}{2}(u_1 - u_0).$$

This is descriptive accounting, not causal attribution. It separates change volume from selected decision quality without claiming either component is a new theorem.

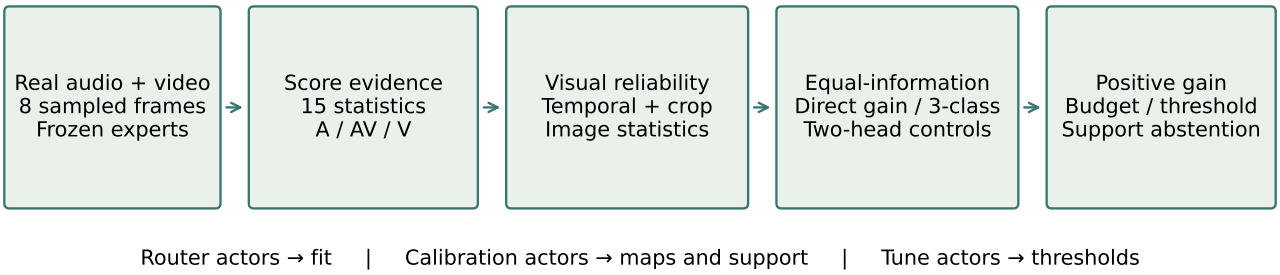


Figure 1. Executed information flow. The visual probes are shared by direct gain, single-head outcome, and structured controls. All expert inference occurs before routing; no encoder saving is claimed.

3. Positive-gain routing with visual support

3.1. Observable evidence

The original fifteen statistics standardize each expert's class logits within a clip, apply softmax, and summarize confidence, margin, entropy, pairwise Jensen–Shannon divergence, and top-one agreement. They use outputs in audio, fusion, video order. True classes, corruption names, actor IDs, and test annotations are never selector inputs.

The new eight-dimensional reliability vector uses the video directly. We encode eight uniformly spaced frames with frozen DINOv2. Adjacent-frame cosine distances contribute their mean and maximum. Four alternate frames are additionally cropped to the central 90% region and resized; original-versus-crop feature distances contribute another mean and maximum. The central 112-by-112 image region supplies luminance standard deviation, mean spatial gradient, mean absolute temporal difference, and mean luminance. These measurements are observable without a clean reference image. They are deliberately modest probes, not proof that a visible object or an emotion is correctly understood.

3.2. Models and equal-information controls

Direct histogram gradient boosting predicts signed gain using eight audio/fusion statistics, all fifteen score statistics, or all twenty-three score and reliability statistics. A single-head classifier predicts rescue, neutral, and harm; its score is the predicted rescue probability minus predicted harm probability. This tests an outcome parameterization equivalent in expressive purpose to conditional heads.

Structured controls estimate $q = \Pr(\Delta \neq 0 \mid Z, D)$ and $r = \Pr(\Delta = 1 \mid Z, D, \Delta \neq 0)$ and score $q(2r - 1)$. We compare raw probabilities, two isotonic maps, and logistic sigmoid maps of clipped probability logits. All new learned methods use the same router actors, three training conditions, and fixed model capacity. The direct and structured models receive identical twenty-three-dimensional information when that comparison is made. The product identity itself is elementary and carries no architectural novelty claim.

3.3. Support-aware abstention

For reliability vector h, compute $d(h) = \sum_j ((h_j - \mu_j)/\sigma_j)^2$, with moments from clean router actors and a standard-deviation floor. The support limit is the 95th percentile of clean calibration distances. The proposed practical rule keeps audio outside that support; inside, it applies the direct twenty-three-feature gain score. This can reject a genuinely useful but unfamiliar video, and can accept a familiar-looking corrupted one. Its value is empirical and must be judged jointly with coverage.

The corrected certificate uses a separate support cutoff estimated only from clean router actors, then divides positive gain scores at their router-derived median. Calibration actors evaluate these fixed regions; they do not define them. A bounded source-mean concentration penalty is subtracted from each region's unconditional gain. Only a positive lower endpoint permits switching. It certifies no regions here; this result prevents the empirical support gate from being mislabeled a validated safety guarantee.

4. Locked protocol and inference

4.1. Data roles and provenance

The inspected CREMA-D test contains 820 clips from ten actors. Its outcomes motivated this study and all its new results are exploratory. No unused CREMA-D actor pool remains in the prior experiment, so we do not invent a fresh within-dataset test. The new routers use the original router, calibration, and tuning roles, nine actors each. For each actor, twenty-four eligible clips are selected by a fixed filename hash. Clean video, a central 25% occluder, and Gaussian blur with standard deviation two define the training-side conditions. All three roles are source-separated; no original test labels fit the new rules.

External confirmation uses the official RAVDESS speech release: twenty-four actors, six shared intended-emotion classes, and four predetermined full-audio–video clips per actor and class, yielding 576 clips. Calm and surprise are excluded before inference because the frozen heads have no corresponding class. Filenames are ranked by a fixed hash within each actor/class stratum. All twenty-four actors are held for external evaluation: there is no RAVDESS calibration, threshold selection, or expert training.

The protocol, fitted selectors, thresholds, and execution code are hashed before RAVDESS feature inference. These are local dated locks, not a claim of third-party registered preregistration. Official ZIP members are retrieved through byte ranges and checked by ZIP CRC; local SHA-256 values are archived. A pre-inference deviation records this bandwidth-efficient retrieval in place of whole-archive MD5 validation. Selection is unchanged. Unknown encoder-pretraining overlap is not ruled out by actor separation.

4.2. Experts, perturbations, and policies

Frozen WavLM Base audio features and DINOv2 ViT-B/14 visual features feed the existing CREMA-trained audio, interaction-fusion, and video heads. Seed 11 is primary; seed 22 repeats the procedure. This measures checkpoint sensitivity conditional on the chosen pretrained encoders, not independent architectural replication.

External conditions are clean video, a static gray square occupying approximately half the letterboxed frame, and a separately specified moving striped occluder occupying approximately 35%. Both corruptions are applied to decoded pixels before feature extraction. The moving intervention is not used in router training. It remains synthetic, and does not substitute for natural occlusion collection.

Three policies remain distinct. A positive-score batch policy ranks only disagreements with positive estimated gain and stops at an at-most budget. Independent thresholds are selected on clean tuning actors with whole-tie handling, clamped to be nonnegative, and fixed under shift. Support-aware independent decisions add the visual-support rejection. Nominal budgets are 5, 10, 20, 30, and 40%; actual external switch rates are always reported.

4.3. Primary comparison and statistics

The primary endpoint is the seed-11 clean RAVDESS paired gain of support-aware fixed-threshold routing over the margin gate at the nominal 20% operating point. A 95% actor interval above zero meets the local primary rule; the equal-information direct-gain comparator is separately necessary for the broader method claim. Other conditions, budgets, and checkpoint results are secondary.

We use 5,000 paired actor bootstrap draws. Independent-threshold masks stay fixed; batch policies are re-ranked within each newly formed bootstrap batch, including repeated actor copies. Inference is conditional on frozen experts and selectors. A planning calculation using earlier paired actor variability yields an approximate 2.37-point minimum detectable effect at twenty-four actors, or about thirty-four actors for two points under the same normal approximation. Domain transfer can change that variance. Twenty-four clusters are an improvement over ten, not an exact tail-probability guarantee.

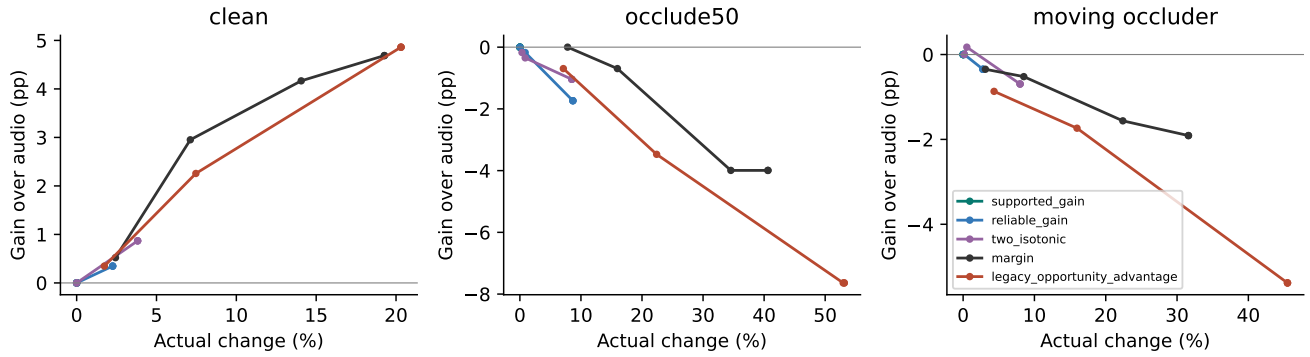


Figure 2. External RAVDESS fixed-threshold frontiers, seed 11. The horizontal axis is the actual change rate, not the tuning budget. Each point retains its clean-tuned threshold under shift.

5. External confirmation results

Policy	Acc.	Gain	Change	Harm
audio	34.38	0.00	0.00	0.00
fusion	42.71	8.33	37.85	6.60
margin	38.54	4.17	14.06	1.39
scores_gain	37.50	3.12	9.03	0.35
reliable_gain	34.72	0.35	2.26	0.00
supported_gain	34.38	0.00	0.00	0.00

At the locked primary operating point, support-aware routing attains 34.38% accuracy and changes 0.00% of predictions. Its gain over audio is 0.00 points, compared with 4.17 for the margin gate and 0.35 for the equal-information direct learner. The paired candidate-minus-margin difference is -4.17 points, with interval [-6.42, -1.91]. Two individually positive gain estimates would not establish this comparison; the paired contrast is the inferential object.

The primary interval is not wholly positive. We therefore retain the failed superiority finding; neither a different budget nor a favorable secondary condition replaces this endpoint.

Coverage and the fallback baseline. Keeping audio everywhere has zero routing gain and zero switch-induced harm. A detector that rejects every corrupted clip can match this endpoint without solving classification under occlusion. Its value must therefore be assessed by both retained clean opportunities and avoided harmful changes. The support fraction, actual change volume, rescues, harms, and neutral switches are supplied for every method and condition. A difference caused solely by nearly universal abstention is not described as superior multimodal reasoning.

Information versus factorization. Comparing the eight-, fifteen-, and twenty-three-feature direct models isolates the value of added observations within a fixed learner family. Comparing direct gain, three-outcome classification, and conditional heads at twenty-three dimensions tests the decision model. An improvement shared by all heads would support useful input information, not a unique two-head architecture. Conversely, a calibration ablation can improve ranking while changing the independent threshold's operating point; its gain and switch volume must be read together.

Complete reporting. The released tables include audio, fusion, and video endpoints; raw and calibrated outcome scores; all budgets; original learned controls where available; and actor-level gains. Captured oracle fractions retain the audio-versus-fusion rescue denominator. Endpoints outside this pair are reported as accuracy controls rather than being silently added to the routing oracle. We retain failures and sources with negative gain rather than selecting a favorable subset for the headline.

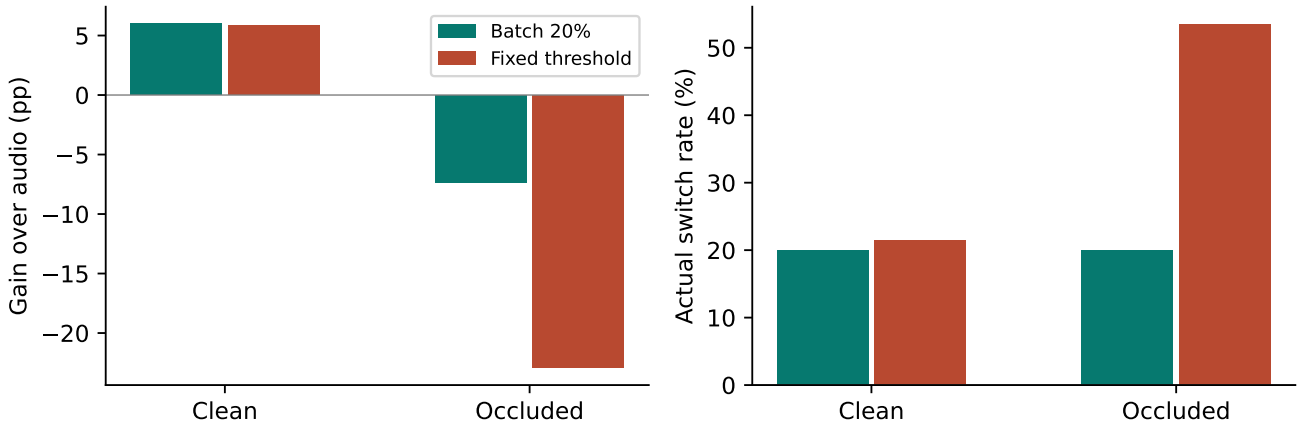


Figure 3. The original CREMA operating failure. A batch quota limits change volume, but its estimate is not the deployment performance of an independent clean-tuned threshold.

6. Why the inspected policy fails

6.1. Positive scores do not guarantee useful switches

The corrected at-most rule is necessary: it never fills a budget with predicted negative gains. At the original 20% operating point, however, it leaves the selected CREMA examples unchanged. Every previously selected score is already positive under clean and occluded inputs. The corrected batch therefore still loses 7.32 points, and the corrected independent threshold still loses 22.93 points under occlusion. The defect is not merely selecting low-ranked negative scores to satisfy a quota.

Across the same clips, the original independent threshold's switch rate grows from 21.46% to 53.41%. Its gain moves from +5.85 to -22.93 points. The symmetric volume/quality identity assigns -2.50 points of that 28.78-point deterioration to the volume term and -26.28 to the selected-quality term. These terms are arithmetic comparisons, not an intervention identifying causality. Nonetheless they rule out describing switch inflation alone as a complete explanation.

The mean selector score increases by 0.119 under the visual intervention, despite its poorer correction outcomes; the exploratory paired actor interval is approximately [0.018, 0.223]. The important inversion is therefore operational: the score moves in a more fusion-favorable direction while selected outcomes become less favorable. The artifact records keep-to-switch, switch-to-keep, retained-switch, and retained-audio transitions, with rescue/harm counts before and after corruption.

6.2. A zero plateau with insufficient support

On the clean inspected test, thirteen disagreements fall in the lowest predicted-rescue bin and eight are rescues. Twelve have conditional rescue probability mapped to exactly zero by isotonic calibration. None of these thirteen raw conditional probabilities lies outside the fitted calibration range. The cause is thus not out-of-range clipping in this group. Six non-neutral calibration clips, drawn from four actors and containing no rescues, support the zero plateau. The held-out cases expose how a locally empty outcome category can become an apparently certain zero after flexible calibration.

This diagnosis does not license fitting a new map to those thirteen test labels. The new comparison fits raw, isotonic, and sigmoid alternatives using the original calibration role and evaluates decision quality independently. Calibration bins, selected-region estimates, and clip identifiers are retained so the claim can be checked against the actual fitted map rather than an aggregate diagram alone.

Exploratory remedy. On the inspected CREMA actors, the new seed-11 support-aware fixed rule gains 4.51 clean points, below the margin rule's 6.59. It rejects every severe gray-occlusion input, yielding zero gain and harm there. The detector recognizes this constructed degradation on CREMA-D, but also rejects all clean RAVDESS videos. Thus the apparent protection fails to distinguish corrupted evidence from benign domain change. The threshold remains unchanged; a post-test relaxation would be development, not confirmation.

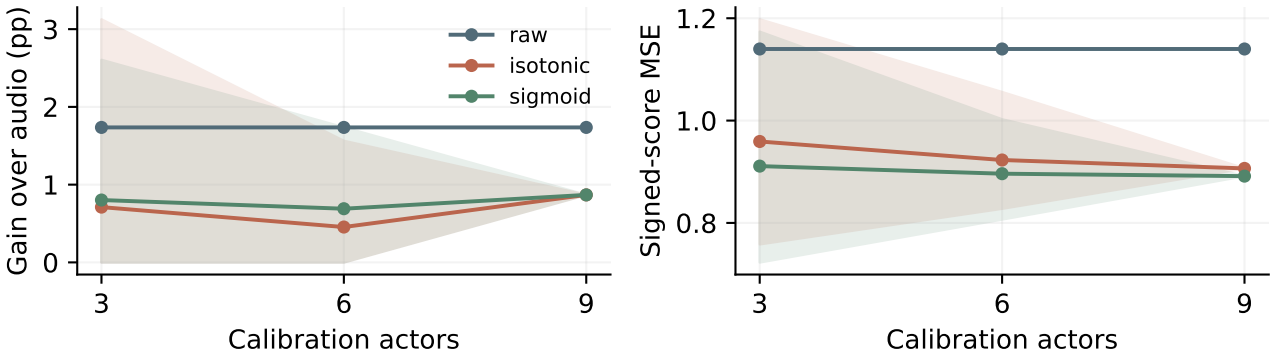


Figure 4. Exploratory actor-subset sensitivity, clean RAVDESS, seed 11. Curves average all calibration subsets of each size; shading is the full subset range, not a confidence interval. Better signed-score error need not improve frozen-threshold decisions.

7. Calibration error versus decision value

7.1. Exhaustive actor-subset sensitivity

The external result was already inspected when this extension was designed. It is exploratory, not a second locked confirmation. We freeze both trained heads and enumerate all 84 three-actor subsets, all 84 six-actor subsets, and the full nine-actor calibration set. For each checkpoint and subset, we fit isotonic and sigmoid calibration to the same source-weighted outcomes; the raw heads supply a repeated fixed reference. The original clean tuning actors choose a nonnegative threshold at an at-most 20% nominal switch budget. No external labels choose calibrators or thresholds.

Five archived evaluation conditions per checkpoint yield 5,070 metric rows. They reuse ten CREMA test actors and twenty-four RAVDESS actors, rather than creating thousands of independent experiments. We report signed-score MSE and rescue/harm Brier error on disagreements, actual coverage, harmful changes, and bias within the selected decision region. Subset ranges measure sensitivity to calibration composition, not population uncertainty.

7.2. An observed mismatch, not a universal inversion

With all nine actors and checkpoint 11, sigmoid calibration reduces clean RAVDESS signed-score MSE from 1.140 to 0.891, while gain drops from 1.74 to 0.87 points and switching from 5.73% to 3.12%. Isotonic calibration gives MSE 0.907 and gain 0.87 points. For checkpoint 22, sigmoid also reduces MSE (0.795 to 0.734), while gain changes only from 5.73 to 5.56 points. These are different operating points, not matched-coverage superiority tests.

The mean loss weights all disagreements; a routing decision depends on errors near an operational boundary and on which positive gains survive it. Improving the former does not imply improving the latter. This is a known decision-theoretic distinction, not a new theorem. Flexible calibration also has established small-sample risks [41]; our contribution here is an executed sensitivity audit of this particular failure.

Across three-actor subsets, checkpoint-11 isotonic gains range from 0.00 to 3.12 points; sigmoid ranges from 0.00 to 2.60. Checkpoint 22 spans 0.17–8.51 and 0.00–7.99 points respectively. Reporting the best subset would hide this instability. All subsets and unsupported constant-class fallbacks are retained.

7.3. Scope of replication

The two checkpoints share their encoders, architecture, source roles, and evaluation actors. They are not independent architectural replications. Score–outcome correlations also do not support a universal inversion: on clean RAVDESS the direct reliability score has Spearman correlation -0.060 for checkpoint 11 but $+0.241$ for checkpoint 22. Raw correlations describe inspected samples and do not carry a significance claim. Physical-object persistence and edited speaker ownership are separate pilots, excluded from evidence for this routing paper.

8. Limits, guarantees, and reproducibility

8.1. What can be guaranteed

The earlier implementation estimated its support cutoff from calibration features and then reused those actors in a fixed-region bound. Absence of outcome-label use does not make that region independent of the sample. That justification is withdrawn. The empirical support-policy results remain unchanged; the repaired certificate is a separate procedure.

Conditional on router training data, fix two disjoint regions using the router-fitted gain head, router-only support cutoff, and router-only score boundary. For each independent calibration actor define Y_{gj} as mean signed gain, including zero outside region j. Since $Y_{gj} \in [-1, 1]$, Hoeffding and a union bound give simultaneous lower endpoints

$$L_j = \bar{Y}_j - \sqrt{2\log(2/\alpha)/n}. \tag{1}$$

With probability at least $1 - \alpha$, both expected region contributions exceed their endpoints. Selecting the union of regions with $L_j > 0$ preserves nonnegative expected gain for the same actor population. Additional filtering inside a certified region is not covered. Calibration must remain independent of every quantity defining these regions, including feature-derived cutoffs.

The repaired cutoff is 20.534, versus the practical policy's 29.981. For nine independent actors and $\alpha = .05$, the penalty is 0.9054. Region means are 0.0278/0.0185 for checkpoint 11 and 0.0278/0.0463 for checkpoint 22: none certifies. A mutation test changes calibration features and labels and verifies that region definitions remain fixed. This tests independence of code inputs, not the population assumptions. No arbitrary-shift guarantee follows.

8.2. Data and target restrictions

Both datasets consist of performed expressions. The target is the intended label, not an inference about a person's internal emotional state. The RAVDESS subset removes classes not represented by the frozen CREMA heads; this is a predetermined six-class transfer evaluation, not a score on the full original eight-class benchmark. The clean external domain differs in actors and recording conditions, but the visual degradations remain constructed. Natural occlusion and a newly collected CREMA-like source population are still missing.

Local media hashes establish file identity, not freedom from unknown pretraining contamination or all semantic near duplicates. The download used official archive ranges, member CRC checks, and local SHA hashes; it does not claim the entire ZIP was downloaded and MD5-verified. The stored protocol deviation records that distinction before external inference. The original CREMA outcomes and VGGSounder exploration were already visible, and cannot supply additional confirmation for the new method.

8.3. Compute and artifact

All selectors are post-fusion: audio, fusion, and video outputs are already available. Reliability adds four crop encodings to the eight video-frame encodings. The release separates audio-encoder, visual-encoder-with-probe, expert-head, and selector timings. Batch timings are not streaming latency or an equal-total-cost frontier. We make no encoder-saving claim.

The archive contains the protocol and parameter locks, source manifests, exact selection hashes, recorded logits and reliability measurements, fitted models, policy masks, all results, paired-bootstrap code, calibration cases, and demonstrations. Portable replay needs no raw media or GPU; full extraction additionally needs the official media and pinned encoders. New test results are now inspected. Any revised model selected from these outcomes needs another independent reserve rather than a renamed split of the same test.

9. Conclusion

The study turns a concealed deployment failure into an explicit comparison of switch volume, correction quality, calibration support, and visual evidence. Positive-score budgeting repairs the decision rule but does not repair confidently wrong gains. Flexible calibration can erase rescue probability in a sparsely supported region. Reliability-aware abstention must be evaluated against direct learners and the audio fallback, with its lost opportunities reported alongside avoided harm. The locked external results, rather than the appearance of a demonstration or the size of an oracle gap, determine the claim supported by this manuscript.

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